

Team Root Matrix: Mary Ann Mastri, Cesar Noriega, Luiz Paludo, Nichelle Graf
A05: "Arrival" Through NLP Lens – Team Contributions
ITAI 2376 – Deep Learning in AI
Professor Viswanatha Rao

L05 - Analyzing "Arrival" Through the Lens of NLP Team Contributions

Mary Ann:

I contributed the initial plot summary and analysis connecting *Arrival* to NLP concepts, including rule-based approaches, representation learning, linguistic relativity, ambiguity, and the “weapon” versus “tool” interpretation. I also provided research sources supporting the discussion of NLP and language structure.

Cesar:

I contributed an analysis of Louise’s step-by-step communication process with the Heptapods, including the use of simple words, visual information, pattern recognition, and context. I also contributed reflections on the difficulty of learning an unfamiliar language without a dictionary or translated examples and on the importance of observation and context in NLP.

Luiz:

I contributed analysis of ambiguity, context, cultural and language differences, and misinterpretation in the movie. I connected these issues to real-world NLP challenges and included personal examples from learning English and working with Portuguese-language sentiment analysis. I also provided supporting references related to NLP and linguistic analysis of *Arrival*.

Nichelle:

I contributed detailed scene-by-scene analysis while watching *Arrival*, identifying connections to semantic grounding, multimodal communication, feature extraction, representation learning, ambiguity, pragmatics, nonlinear language, and linguistic relativity. I also developed an original reflection on how the communication process in *Arrival* might differ if the story were made today, especially given the development of Transformers and modern multimodal AI since the film’s 2016 release. My analysis was already moving toward the role of speech analysis, computer vision, tokenization and language pipelines, contextual models, and human evaluation, although Professor Rao later identified those five technology areas explicitly in his submission comments. I also integrated the individual team submissions into the final written report, organized the content around the assignment rubric, refined the group analysis for clarity and coherence, created GitHub repository and prepared supporting repo documentation.